

Advanced AC Circuit

Laboratory Exercise No: 4

Measurement for Active RC Filter

Introduction

Active filters consist of combinations of resistors, capacitors, and op amps. They offer some advantages over passive RLC filters. Firstly, they are often smaller and less expensive, because they do not require inductors. This makes feasible the integrated circuit realizations of filters. Secondly, they can provide amplifier gain in addition to providing the same frequency response as RLC filters. Thirdly, active filters can be combined with buffer amplifiers (voltage followers) to isolate each stage of the filter from source and load impedance effects. This isolation allows designing the stages independently and then cascading them to realize the desired transfer function. (Bode plots, being logarithmic, may be added when transfer functions are cascaded.) However, active filters are less reliable and less stable. The practical limit of most active filters is about 100 kHz—most active filters operate well below that frequency.

In this lab you will learn to measure the magnitude response and phase response of active low-pass, high-pass and band-pass filter and characterize their properties.

1. Active RC low-pass, high-pass and band-pass filters

- Build the circuit shown in Fig.1 (a) and (b), respectively. Use **Function Waveform Generator** as voltage input v_s and use **Digital Oscilloscope** to observe and measure the value of output v_o and phase difference between output and input at the selected frequencies.

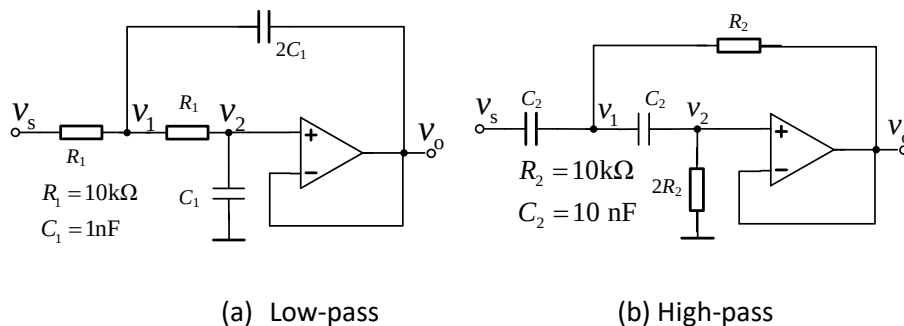


Figure 1: Active RC Filters

- Cascade the circuits shown in Fig.1(a) and (b) to form a band-pass filter. Measure its magnitude response and phase response.

2. Pre-Lab Work

a) Theory analysis

- Use phasor analysis method to determine the transfer function — voltage gain for the circuits shown in Fig.1(a) and (b), respectively. And then determine the transfer function — voltage gain for the band-pass filter you build. Calculate the amplitude of these transfer functions and their phase differences between the input and output voltages.
- Express all these amplitude responses in dB.

b) Simulation Practice

Use Multisim 14.0 to simulate your circuit. Make use of a **Bode Plotter** to observe frequency response. Use cursor to find the amplitude and phase of the transfer function for low-pass, high-pass and band-pass at frequencies given in **Table 1**, **2** and **3**, correspondingly. Output the Bode plots of each transfer function.

Table 1. Data for low-pass filter

Source frequency/kHz	0.1	1	10	50	100	150
$ \dot{V}_o / \dot{V}_s / \text{dB}$						
$\Delta t / \mu\text{s}$						
φ / Deg						

Table 2. Data for high-pass filter

Source frequency/kHz	0.1	0.5	1	10	100	150
$ \dot{V}_o / \dot{V}_s / \text{dB}$						
$\Delta t / \mu\text{s}$						
φ / Deg						

Table 3. Data for band-pass filter

Source frequency/kHz	0.1	0.5	1	4	7	10	50	100	150
$ \dot{V}_o / \dot{V}_s / \text{dB}$									
$\Delta t / \mu\text{s}$									
φ / Deg									

Table 4. Data for -3dB frequencies

$f_{\text{LP-3dB}}$	$\varphi_{\text{LP-3dB}}$	$f_{\text{HP-3dB}}$	$\varphi_{\text{HP-3dB}}$	$f_{\text{L-3dB}}$	$\varphi_{\text{L-3dB}}$	$f_{\text{H-3dB}}$	$\varphi_{\text{H-3dB}}$

3. Lab Section 1

In this laboratory, you are required to use **Digital Multi-meter** to measure voltage and to use **Oscilloscope** to measure phase. Build the circuit shown in Fig.1 on a bread board. Use **Function Generator** as input. Set v_s rms to 0 dB and set the frequency given in Table 1, 2, and 3. Observe waveforms on the oscilloscope and measure the phase difference. Use Digital Multi-meter to measure the output voltage. The procedure includes the following steps:

- (i) Measure the resistance and capacitance rated $R=10\text{ k}\Omega$, $C=1\text{ nF}, 10\text{ nF}$.
- (ii) Build the circuit shown in Fig.1(a) and then (b) on breadboard and connect the instruments correctly. **Make sure to supply power to op amp.**
- (iii) Select Sine Function from **Function Waveform Generator** as v_s , and monitor it on the **Digital Oscilloscope** (CH1). Use CH2 to monitor output voltage v_o . Use **Digital Multi-meter** to measure voltage.
- (iv) Observe the frequency response roughly to see if you have done it correctly. Set a low frequency on the source and increase it. Adjust **Digital Oscilloscope** to show stable waveforms at lower frequency and then higher frequency to see if it performs the property of low-pass or high-pass.
- (v) Measure the voltages and phases and fill the results in Table 1, 2, and 3. Compare the results with that from simulation.

Requirements and notes

- (i) While measuring, every time you change the frequency, make sure to adjust v_s so that the measuring result of v_s keeps unchanged.
- (ii) Use double traces method to measure phase difference at each frequency point.
- (iii) Fill all the results in the data tables.
- (iv) Determine -3dB frequency f_{LP-3dB} and phase deference ϕ_{LP-3dB} in Fig.1(a);
Determine -3dB frequency f_{HP-3dB} and phase deference ϕ_{HP-3dB} in Fig.1(b);
Determine -3dB frequencies at lower and higher side f_{L-3dB} , f_{H-3dB} and phase deference ϕ_{L-3dB} , ϕ_{H-3dB} in band-pass.
Fill the above results in Table 4.
- (v) Sketch semilog magnitude response and phase response graphs for each filter.
- (vi) From the magnitude response, conclude the frequency selectivity for each filter.

(vii) Compare the main difference in magnitude response and phase response between 1st-order and 2nd-order filters.

4. Lab Section 2

In this laboratory, you will learn how to design a band-pass filter. The procedure includes the following steps:

(a) Make use of Fig.1 to design a band-pass filter. Select proper values of resistors and capacitors such that -3dB frequencies are at 3kHz and 30kHz.

(b) Assess your design by theory analysis along with simulation or practical test.

5. Submission

a) Complete the theory analysis including equation deductions;

b) Complete the software simulation including the simulating circuits, magnitude and phase response plots, measurement results at each frequency, and output waveforms;

c) Complete the practical measurements including data filling in forms, waveforms output and frequency response plots (in semilogarithm);

d) Complete the analysis as required in Lab Section1.

e) Make sure the names and IDs of **BOTH** team members are on the report,

f) You should submit the completed report to Moodle on time.

6. Assessment

i) This lab is assessed by 100marks.

ii) The prelab work is assessed individually by 30% which includes theory analysis and simulation work. **Submit the prelab paper to Moodle before you enter the lab.**

iii) The lab work is designed to be completed in groups of two. Each group is required to submit ONE report which is assessed by 70%. Prelab work should be still a part of report.